Abstract Linear Algebra

Prof. Lanier

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Pick up a handout. Try those warm-up problems.

Next up:

- This week: finish Ch 5; Ch 6.1-3 and Kalman article for SVD
- HW6 (written and WeBWorK) due tonight.
- HW6 debrief/prebrief today, 3:30-4:30 in Eckhart 207A;
  No other office hours today.
- Problem swap 2: solutions due on Wednesday night
- Quiz 10 on Friday (orthogonality, Gram–Schmidt)
- Problem session / office hours on Friday, 3-5pm, Ryerson 177
- HW7 due next Monday
- Make a move on flex!


Many matrix decompositions

QR decomposition: Let A be an invertible matrix in an inner product space. Then $A = QR$ where Q has orthonormal columns and R is upper-triangular with positive diagonal entries. [Gram–Schmidt]

Upper triangular (Schur) representation (Thms. 6.1.1 & 6.1.2): Let A be the matrix of an operator on an inner product space. Then A can be represented as $A = UTU^*$, where U is a unitary matrix and T is an upper triangular matrix. In the case of real square matrices, any real square matrix A with all real eigenvalues can be represented as $T = UTU^{-1}$, where U is a real orthogonal matrix and T is a real upper triangular matrix.

Spectral Theorem (Theorem 6.2.1): Let $A = A^*$ be a self-adjoint operator in an inner product space X (the space can be complex or real). Then all eigenvalues of A are real, and there exists and orthonormal basis of eigenvectors of A in X. In the real case, this says that symmetric matrices ($A = A^T$) are diagonalizable with an orthonormal change of basis matrix: $A = QDQ^{-1}$

Singular Value Decomposition (Def 6.3.9, see Kalman article for proof): Every matrix A has a singular value decomposition $A = W\Sigma V^*$, where W and V are unitary (orthogonal for $\mathbb{R}$) matrices and Σ is "diagonal" with real numbers on the beginning of the diagonal and 0's elsewhere.

D+N (Cor. 9.3.9): Any operator A in V can be represented as $A = D + N$, where D is diagonalizable (i.e. diagonal in some basis) and N is nilpotent ($N^m = 0$ for some m), and $DN = ND$.

Jordan decomposition theorem (Thm. 9.5.1): Given an operator T there exist a basis (Jordan canonical basis) such that the matrix A of T in this basis has a block diagonal form with blocks of form

$$\begin{pmatrix} \lambda & 1 & & & 0 \\ & \lambda & 1 & & \\ & & \lambda & \ddots & \\ & & & \ddots & 1 \\ 0 & & & & \lambda \end{pmatrix}$$

where λ is an eigenvalue of A. (Here we assume that the block of size 1 is just λ.) The matrix A is unique up to permutation of the blocks.

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where λ is an eigenvalue of A. (Here we assume that the block of size 1 is just λ.) The matrix A is unique up to permutation of the blocks.

Cayley–Hamilton theorem

Find a square matrix A that satisfies

$$A^3 = A$$

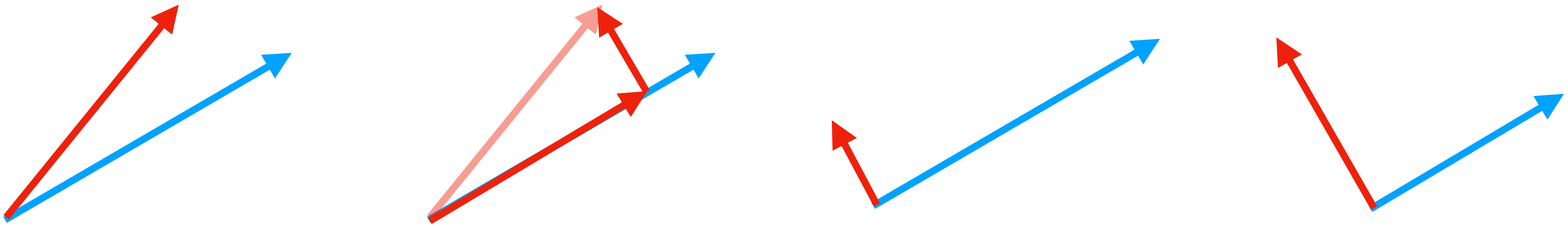
or how about:

$$A^3 - 5A^2 + 6I = 0$$

Gram–Schmidt process:

$$\mathrm{proj}_v(u) = \frac{u \cdot v}{v \cdot v} v$$

Modify vectors in turn to make them orthogonal to the previous basis vectors, and then normalize them to give them unit length.



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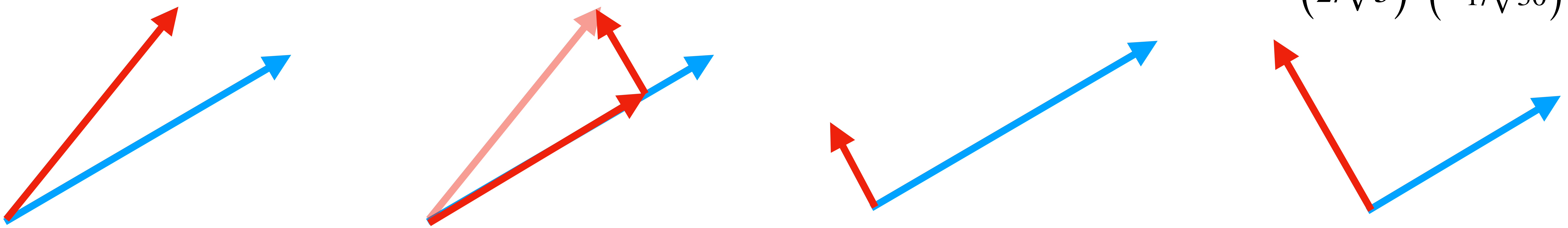
$$\begin{pmatrix} 1/\sqrt{5} \\ 0 \\ 2/\sqrt{5} \end{pmatrix} \begin{pmatrix} 2/\sqrt{30} \\ 5/\sqrt{30} \\ -1/\sqrt{30} \end{pmatrix}$$



$$\begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

$$\mathrm{proj}_v(u) = \frac{1 \cdot 1 + 1 \cdot 0 + 1 \cdot 2}{1 \cdot 1 + 0 \cdot 0 + 2 \cdot 2} \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} = \frac{3}{5} \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix}$$

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$$\begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \end{pmatrix} \quad \begin{pmatrix} 1 \\ 2 \\ 2 \\ 0 \end{pmatrix} \quad \begin{pmatrix} 0 \\ 4.5 \\ 0 \\ -1 \end{pmatrix}$$

span a 3-dimensional subspace of $\mathbb{R}^4$.

Find an orthonormal basis for this subspace.

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Thm. Every subspace of an inner product space has an orthonormal basis.

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Pf. The Gram–Schmidt process.

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(into elementary matrices and LU and PLU)
so does Gram–Schmidt.

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$$\begin{pmatrix} 1 & 2 \\ 1 & 3 \end{pmatrix} = \begin{pmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ -1/\sqrt{2} & 1/\sqrt{2} \end{pmatrix} \begin{pmatrix} \sqrt{2} & \sqrt{2} + 1/3/\sqrt{2} \\ 0 & 1/\sqrt{2} \end{pmatrix} \approx \begin{pmatrix} \cos \pi/4 & \sin \pi/4 \\ -\sin \pi/4 & \cos \pi/4 \end{pmatrix} \begin{pmatrix} 1.41 & 3.53 \\ 0 & .707 \end{pmatrix}$$

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$$A \begin{pmatrix} 3 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

(subtract a multiple of first column from second column to make them orthogonal)

(scale second column to make it length 1)

(...)

$$= Q$$

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$$A \begin{pmatrix} 3 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & -2 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \text{(scale second column to make it length 1)} \quad (\ldots) = Q$$

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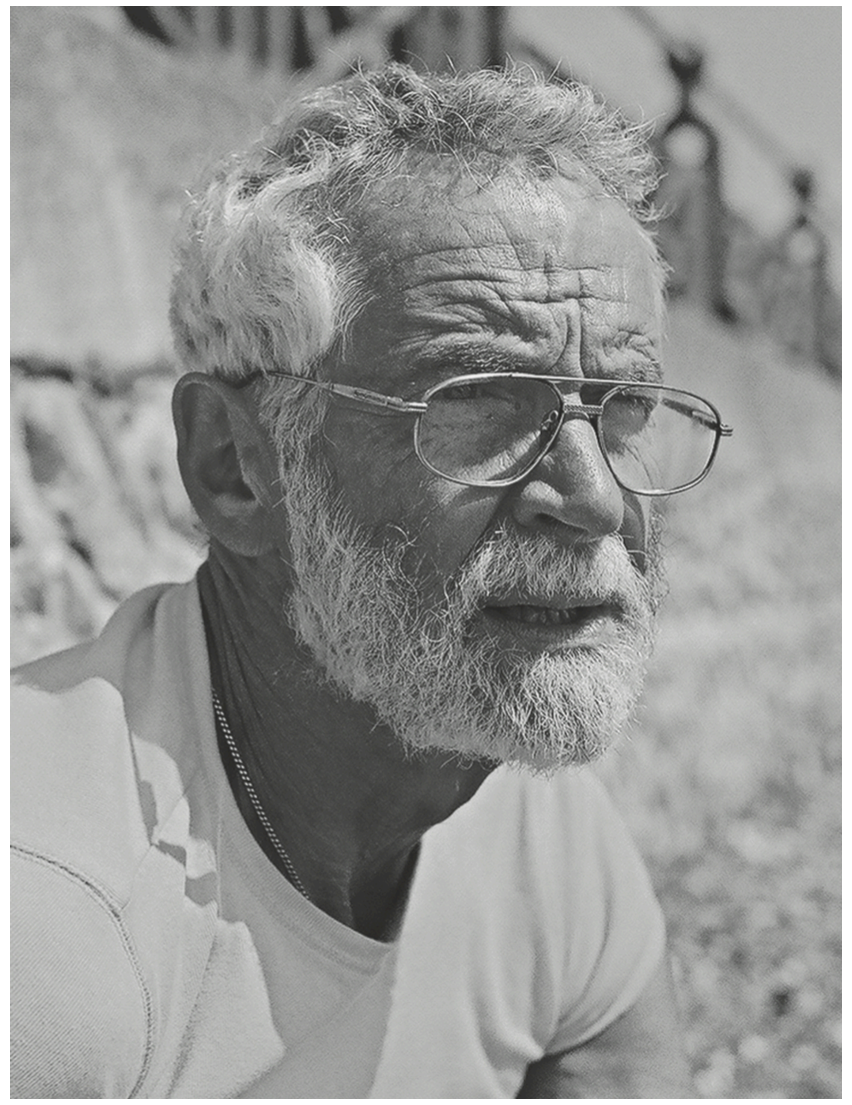
The QR algorithm for computing eigenvalues (1960)

$$A = A_1 = Q_1 R_1$$



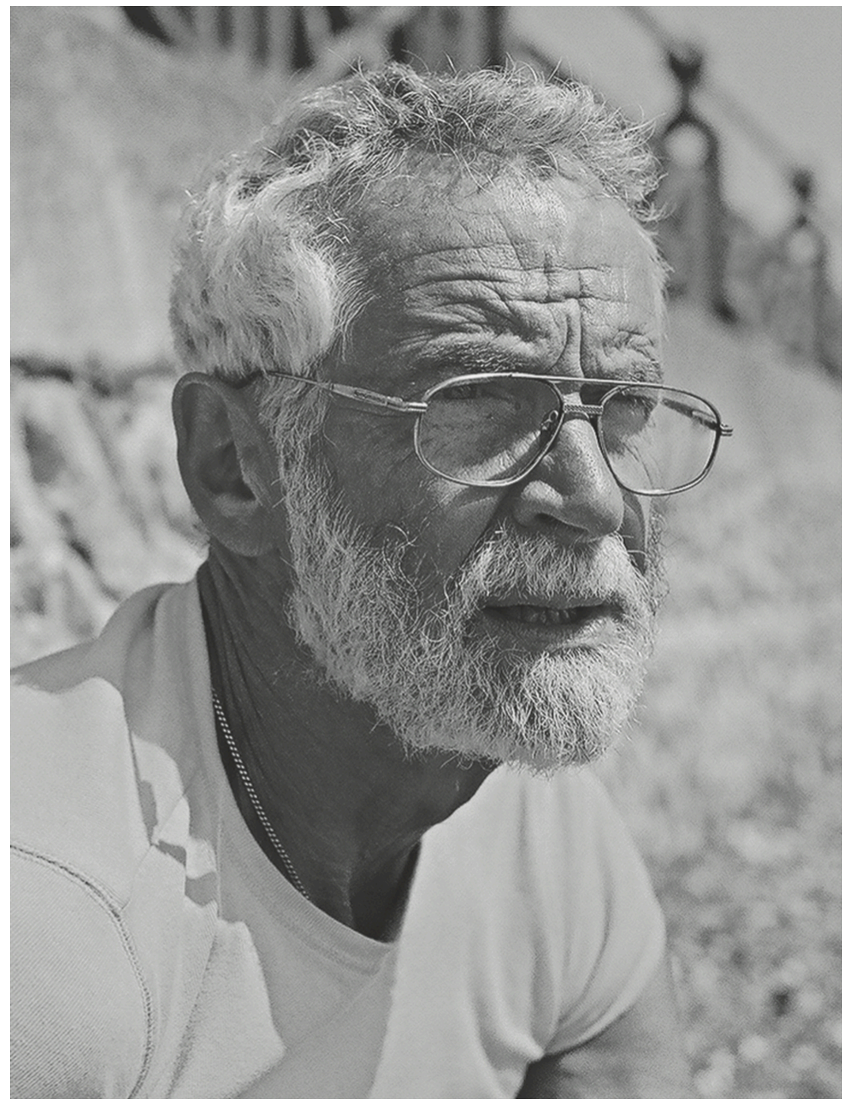
John Francis, 2008

Vera Kublanovskaya, 2008

The QR algorithm for computing eigenvalues (1960)

$$A = A_1 = Q_1 R_1$$

$$R_1 Q_1 = A_2 = Q_2 R_2$$



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The QR algorithm for computing eigenvalues (1960)

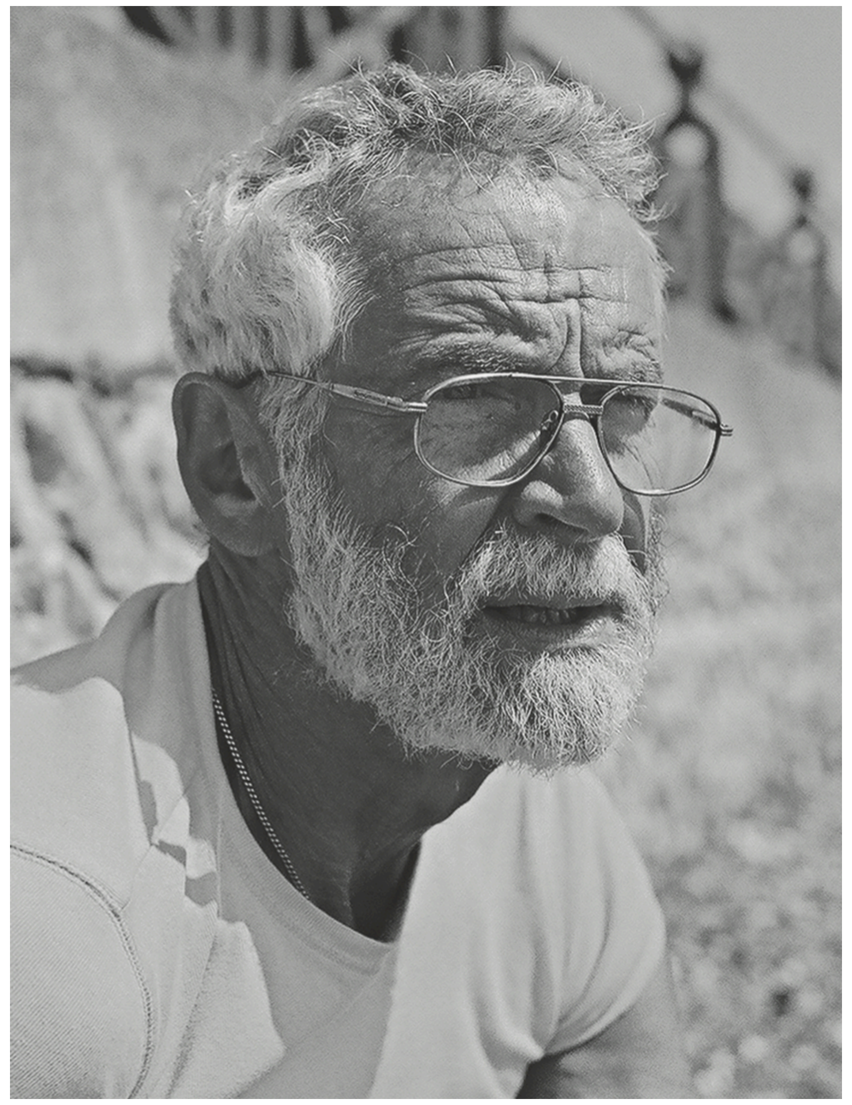
$$A = A_1 = Q_1 R_1$$

$$R_1 Q_1 = A_2 = Q_2 R_2$$

$$R_2 Q_2 = A_3 = Q_3 R_3$$

$$\vdots$$

Under the right conditions, the matrices A_k converge to a triangular matrix, the Schur form of A with the eigenvalues of A on the diagonal.



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WU

7 things (so far!) that row reduction is great for:

- Solving a system of linear equations
- Checking a set of vectors for: linear independence, spanning, basis
- Computing the inverse of an invertible matrix
- Computing the rank and nullity of a matrix
- Finding factorizations of a matrix: into elementary matrices;
  A=LU and A=PLU and A=PLDU factorizations
- Calculating det A
- Finding eigenvalues and eigenvectors of A.

9 things (so far!) that row reduction is great for:

- Solving a system of linear equations
- Checking a set of vectors for: linear independence, spanning, basis
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- Computing the rank and nullity of a matrix
- Finding factorizations of a matrix: into elementary matrices;
  $A=LU$ and $A=PLU$ and $A=PLDU$ and $A=QR$ factorizations
- Calculating $\det A$
- Finding eigenvalues and eigenvectors of A.
- Finding orthonormal bases.
- Finding orthogonal complements to subspaces.

Orthogonality fun facts!

The row space and nullspace of a square matrix A are orthogonal subspaces.
(in fact, orthogonal complements)

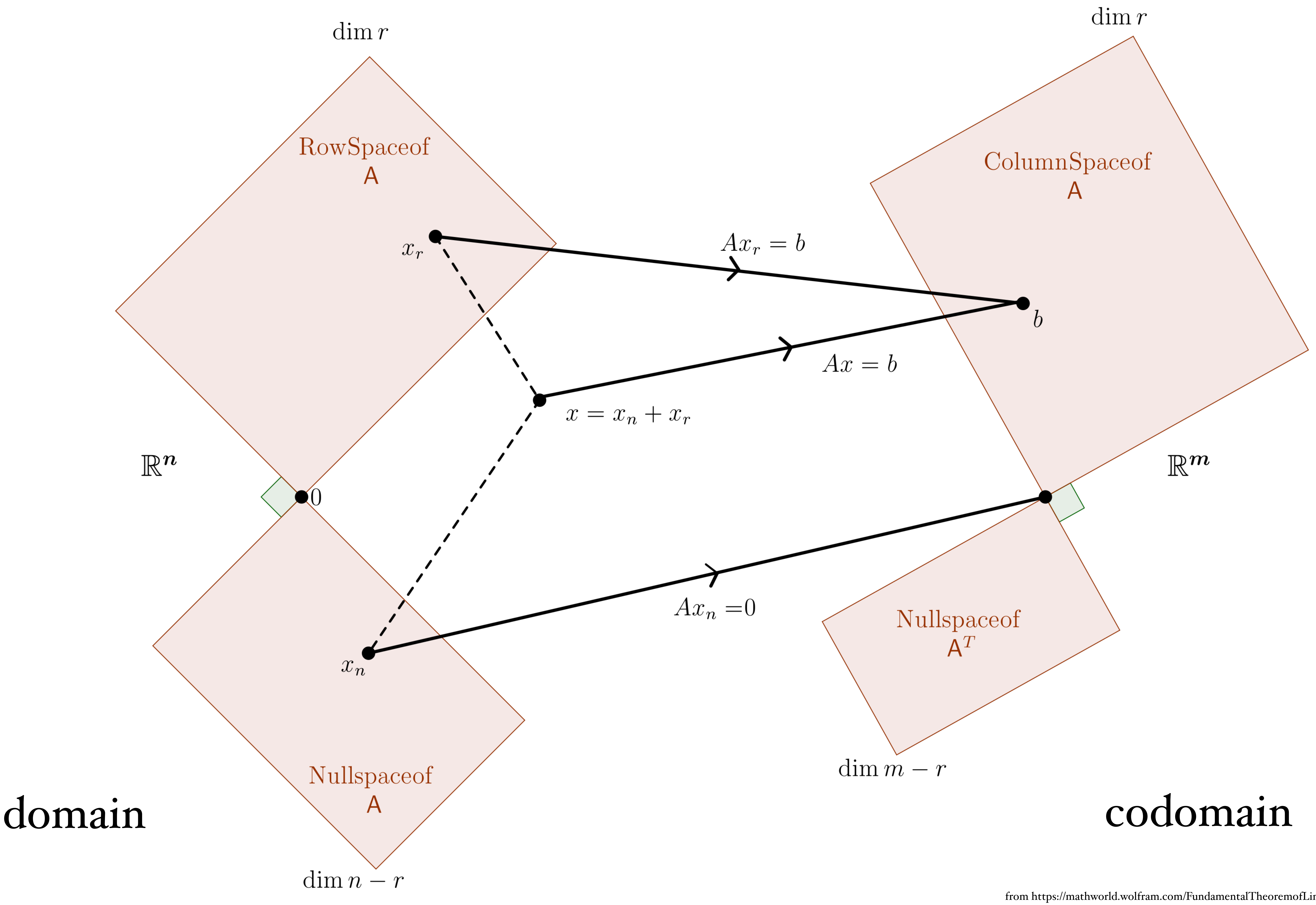
Same thing for the column space and the left null space.

For an orthogonal matrix A (one with columns an orthonormal basis) we have $A^{-1} = A^T$.

The determinant of a real orthogonal matrix A is either 1 or -1.

Note as special cases: every permutation matrix is orthogonal.

The fundamental theorem of linear algebra



from https://mathworld.wolfram.com/FundamentalTheoremofLinearAlgebra.html

Next up:

- This week: finish Ch 5; Ch 6.1-3 and Kalman article for SVD
- HW6 (written and WeBWorK) due tonight.
- HW6 debrief/prebrief today, 3:30-4:30 in Eckhart 207A;
  No other office hours today.
- Problem swap 2: solutions due on Wednesday night
- Quiz 10 on Friday (orthogonality, Gram–Schmidt)
- Problem session / office hours on Friday, 3-5pm, Ryerson 177
- HW7 due next Monday
- Make a move on flex!